ACKNOWLEDGEMENT

We take this opportunity to represent our project on **"IMPACT IMPROVEMENT OF FRONTAL BUMPER PIPE USING PLA MATERIAL AND 3D PRINTING TECHNOLOGY".** We express our sincere thanks to our project guide **Dr. B. K. Suryatal** and co. guide **Prof. A. D. Gaikwad** for their valuable help, guidance and the confidence which they gave us at all stages of the project work. We also thanks to **Dr. S. A. Patil** Head of the Mechanical Department. For providing us the necessary facilities in the laboratory and usage of Internet.

We are thankful to our principal **Dr. R. V. Patil** And to all our staff members who encouraged us to do this project.

Finally we owe our thanks and deep appreciation much more than words can express to our Parents and Friends for their Cooperation, Constant Support and Encouragement without which this would have been a distant dream.

Mr. Sudhir Mali

Mr. Saurabh Tule

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Mr. Rakesh Jadhav

Miss Pournima Nale

Miss Pratiksha Hole

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ABSTRACT

In recent years, the safety improvement of the vehicles is important as well as the engine performance and fuel consumption is also important. The customers are mainly focus on the fuel efficiency and safety rating of the car. There are many cars available in markets which has low safety ratings by global NCAP test. So decide to focus on the frontal impact of the car by improving the shock absorbing capacity of the vehicle.

We can use recycled waste plastic material to improve the impact and increase the shock absorbing capacity of the frontal bumper because of density of the plastic. Plastic material is light weight compare to the metals so that not affect to the vehicle weight and engine performance. Design the different types of 3D printing structure in CATIA by refereeing frontal bumper hollow pipe. Install that plastic structure into hollow pipe and perform the dynamic analysis on the all structure. Find out best structure by checking reaction force of the structure. Perform the UTM testing to validate the results.

Chapter 1

INTRODUCTION

Plastic/Metal Hybrid technology was conceived in the late 1980s. Utilizing our expertise, General Motors developed a prototype instrument panel support structure. Today, automobile companies such as Audi, Ford, and Nissan use LANXESS Plastic/Metal Hybrid technology for the construction of parts ranging from front-end structures to seating and more. It provides strength, rigidity, and energy absorption with weight reduction for fuel efficiency.

The front-end modules were mostly designed and fabricated with heavy steel materials. However, in recent years, lightweight composite materials, like polypropylene, glass mat thermoplastic, long fiber thermoplastic (LFT), direct LFT aluminum, etc. are used in manufacturing front-end modules. The demand for lightweight materials from automakers has significantly increased over the decade, owing to the enactment of stringent emission standards. As lightweight auto parts reduce the overall weight of the vehicles, it can, in turn, reduce CO_2 emissions and cost. Experimenting with front-end module materials in auto parts has predominately contributed to achieving 20%-30% of vehicle weight reduction.

The car industry uses a tremendous number of materials to build cars, including iron, aluminum, steel, glass, rubber, petroleum products, copper, steel and others. These materials have evolved greatly over the decades, becoming more sophisticated, better built, and safer. They've changed as new automotive manufacturing technologies have emerged over the years, and they're used in increasingly innovative ways. This article is devoted to systematization information on the introduction and application of modern materials in the automotive industry.

Composites, as relatively new materials, still do not have a generally accepted definition. Composites are made by compounding of two or more different materials. Their basic constituents have own different characteristics and properties, while the compound presents a completely new material. This material has its own unique, completely new and different properties in relation to constituent components. The aim of this material compounding is to improve structural, tribological, thermal, chemical or some other material properties. Constituent components do not mix and dissolve between each other, so two or more phases are present within composite material.

Using 3D printing plastic in a steel pipe can help in increasing the overall strength of the component while achieving all performance targets. 3D printing or additive manufacturing is a process of making three dimensional solid objects from a digital file. The creation of a 3D printed object is achieved using additive processes. In an additive process an object is created by laying down successive layers of material until the object is created. Each of these layers can be seen as a thinly sliced cross-section of the object. 3D printing is the opposite of subtractive manufacturing which is cutting out / hollowing out a piece of metal or plastic with for instance a milling machine. 3D printing enables you to produce complex shapes using less material than traditional manufacturing methods. Adoption of 3D printing has reached critical mass as those who have yet to integrate additive manufacturing somewhere in their supply chain are now part of an ever-shrinking minority. Where 3D printing was only suitable for prototyping and one-off manufacturing in the early stages, it is now rapidly transforming into a production technology.

Fig.1 Front bumper rectangular pipe

Chapter 2

LITERATURE SURVEY

1. "Crash analysis of a passenger car bumper assembly to improve design for impact test" by Mohammed Abdul Basith, N. Chandrashekar Reddy, Sudhakar Uppalapati, S.P. Jani. 2214-7853/ 2020 Elsevier Ltd.https://doi.org/10.1016/j.matpr.2020.08.561.

In this research paper author mohammed abdul basith et al, explain about the crash analysis of the front bumper. Study on the design of the front bumper to improve the performance in impact test. In this study author mainly study this crash test on the three specific speed which is 40, 60 and 80 km/hr. In the recent years the material of the vehicle front bumper is aluminium alloy thermoplastic and mild steel. The Majority of passenger vehicle accidents occur on the front side. So to absorb the impact force and the energy released during the collision of a vehicle the material and the design of the bumper should be considered to improve safety to the passengers in the car and reduce the damage of the body parts. Now with the improved research in the field of design, more affordable and lightweight materials are available at affordable rates and can be used for manufacturing of a passenger vehicle bumper. The reduced weight of the bumper also increases the fuel efficiency of the car and also stability in performance of the car bumper. The repair cost of the material is affordable, more flexible for modification.

2. "3D printing as an enabling technology to implement maritime plastic Circular Economy" by J. Garrido, J. Saez, J.I. Armesto, A.M. Espada, D. Silva, J. Goikoetxea, A. Arrillaga. Procedia Manufacturing 51 (2020).

In this literature author j Garrido explain about the 3D printing technology. The strategy is a combination of Circular Economy principles, with the use of ocean plastic waste for developing new greener materials, and the uptake of advanced manufacturing technology, 3D printing, flexible enough to adapt to the manufacturing conditions for new Eco-innovative small and medium parts and components. The paper presents the ongoing research in the project about strategies to introduce Circular Economy in the maritime sector from plastic wastes. This paper has presented the results of the research survey about strategies to introduce a Circular Economy in the maritime sector for plastic wastes.

The survey has brought empiric evidence about the economic, technical and environmental viability 3DP technology in the manufacturing of parts using green materials.

3. "Design and manufacture of automotive composite front bumper assemble component considering inter-facial bond characteristics between overmolded chopped glass fiber polypropylene and continuous glass fiber polypropylene composite" by Sung-Jun Jooa , Myeong-Hyeon Yua , Won Seock Kimb , Jong-Wook Leeb , Hak-Sung Kim. https://doi.org/10.1016/j.compstruct.2019.111849 Received 4 March 2019.

In this article author study on the four wheeler front bumper assemble component with the composite material glass fibre. This paper focus on the glass fiber polypropylene composite material for front bumper assembly. This work presents the experimental and finite element analysis (FEA) study of the bond behavior of a continuous glass fiber-reinforced thermoplastic composite rod (3D-Tow)/over-molded long chopped glass fiberreinforced thermoplastic (LFT) composite. To obtain the inter-facial bond strength between the 3D-Tow and LFT of the composite, pull-out tests were performed according to over-molding conditions of the LFT material to investigate the bond-property changes with respect to the process parameters. From the experiment results, the traction–displacement behavior of the 3D-Tow/LFT interface was derived based on the cohesive zone model.

4. "Design improvements of vehicle bumper for low speed impact" by Neraj Natarajan, Preeti Joshi , R.K. Tyagi. https://doi.org/10.1016/j.matpr.2020.08.212 2214-7853/ 2020 Elsevier Ltd. All rights reserved.

In this research paper author R. K. Tyagi study on the vehicle bumper design improvement. The author mainly focus on design for the low speed impact of vehicle. The key role of the bumper is to provide the safety to the expensive parts like engine, radiator, cooling mechanism, headlights etc. The bumper also helps in the energy absorption during high and low speed impacts. Generally, the slow speed impact condition is not considered while designing of a bumper, and the amount of energy absorption is not considered for slow speed. ''In this paper, low speed impact analysis of a car bumper is performed for the central impact and corner impact. Five different design variations are performed using finite element analysis. Parameters like the shape of the bumper and thickness are varied and impact condition is taken into consideration. The fifth design that is ''Effect of increased thickness for modified profile is found to be best in energy absorption during impact.

5. "Large core plastic planar optical splitter fabricated by 3D printing technology" by VáclavPrajzler, Pavel Kulha, Marian Knietel, Herbert Enser. http://dx.doi.org/10.1016/j. optcom.2017.04.070 Received 1 March 2017.

In this research paper author vaclav prajzler et al, study on the Large core plastic planar optical splitter fabricated by 3D printing technology. The splitter was designed by the beam propagation method intended for assembling large core wave guide fibers with 735 µm diameter. Wave guide core layers were made of optically clear liquid adhesive, and Vero-clear polymer was used as substrate and cover layers. Measurement of optical losses proved that the insertion optical loss was lower than 6.8 dB in the visible spectrum. In the paper we reported on our new optical splitter in which the substrate with the U-groove and the upper cladding layers were made by 3D printing technique. The wave guiding layer was made by fillingup the Y-shaped U-grooves with UV-cured optical adhesive polymer and Veroclear polymer as the substrate and cover protection layers was used. The dimensions of the splitter were optimized by beam propagation method using RSoft software.

6. "Metal-plastic hybrid 3D printing using catalyst-loaded filament and electroless plating" by Jing Zhan, Takayuki Tamura , Xiaotong Li , Zhenghao Ma , Michinari Sone, Masahiro Yoshino, Shinjiro Umezu, Hirotaka Sato. https://doi.org/10.1016/j.addma.2020.101556 Received 20 February 2020.

In this article author study on the 3D printing technology with the metal plastic hybrid technology using catalyst loaded filament and electroless plating. This paper proposes and demonstrates a plastic 3D printing technology that adopts electroless plating, a form of chemical metal deposition. The technology is capable of metalizing selected areas of 3D-printed plastic structures made of acrylonitrile butadiene styrene (ABS). Because electroless plating is triggered by a palladium (Pd) catalyst, this study designed and custom fabricated an ABS filament that contains palladium chloride ($PdCl_2$) as a catalyst precursor, which can be used in a fused filament fabrication (FFF) 3D printer.

Chapter 3

PROBLEM STATEMENT

Traditionally, front-end modules consisted of hefty steel carriers. But with the growing focus on reducing the weight of the vehicles in the past few years has led to increasing usage of lightweight composites, instead of steel and iron, for the structural carriers of FEMs. The main 3D printing material used is plastic, although some metals can also be used for 3D printing. However, plastics offer advantages as they are lighter than their metal equivalents. This is particularly important in industries such as automotive and aerospace where light-weighting is an issue and can deliver greater fuel efficiency. Also, parts can be created from tailored materials to provide specific properties such as heat resistance, higher strength or water repellence.

Chapter 4

OBJECTIVES

- Strength improvement of the frontal bumper for impact improvisation.
- 3D modelling of the metal hollow pipe used in vehicle front bumper with different 2 types of plastic structure.
- Perform the dynamic analysis on the hollow pipe.
- Perform the dynamic analysis of the hollow pipe with different plastic structure.
- Study on the 3D printing technology and material properties of PLA.
- Experimental validation of the optimized 3D printing structure with the help of universal testing machine.

Chapter 5

METHODOLOGY

Stage 1

- In this semester we started the work of this project with literature survey.
- Gathered many research papers which are relevant to this topic

Stage 2

- After gathered research paper we describe literature gap on, and identify need of project.
- Then started work on concept design with the help of literature survey market survey.

Stage 3

- After finalizing concept 3D model of project, we started work on material selection design and also discuss about component.
- After deciding the components, the 3D Model and drafting will be done with the help of CATIA software.

Stage 4

- The Analysis of the components will be done with the help of ANSYS using FEA.
- The experimental observations will be taken on compenent.

Stage 5

- Meshing the CAD model and applying the boundary conditions.
- Solve for the solution of meshed model using ANSYS.

Stage 6

- Fabrication of specimen
- Suitable experimentation
- Validation of result by comparing with software results.

Chapter 6

GLOBAL NCAP TEST

Over the past year, Global NCAP has released the results of more models, most recently of the Hyundai Santro, Datsun Redigo and Maruti's Ertiga and Wagon R. While the results have reached far and wide (God bless the internet), a look at online forums and social media chatter also suggests there's a lot of misinformation about the crash tests. So, it's a time as good as any to break it right down.

NCAP stands for New Car Assessment Program. To give a brief background, in 1978 USA became the first country to come up with a programme to provide car crashworthiness information to consumers, which eventually expanded to crash testing and reporting the results. The US-NCAP model formed the basis for similar programme in other regions, and today there is the Australasian NCAP, Euro NCAP, Japan NCAP, ASEAN NCAP, China NCAP, Korean NCAP and Latin NCAP. Global NCAP, an independent charity registered in the UK, was formed in 2011 to enhance cooperation between the various NCAPs and primarily promote vehicle crash-testing and reporting in emerging markets. 'Safer Cars For India' and 'Safer Cars For Africa' are its key initiatives at the moment.

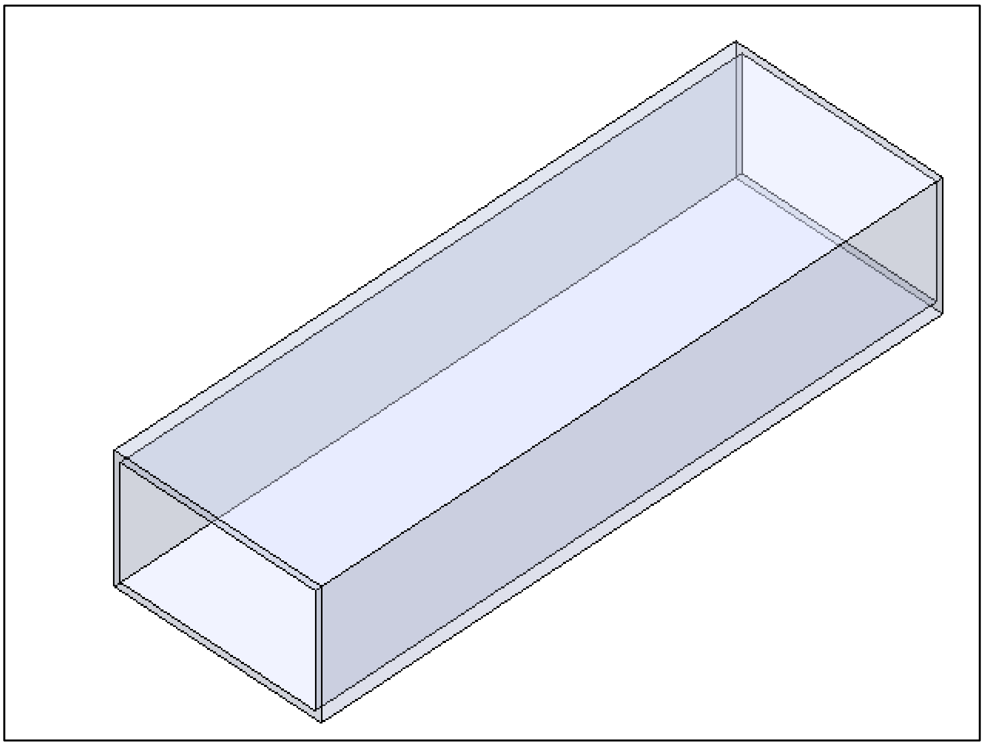
Fig.2 Crash test

RESULTS 2014 - 2019			
Tata Nexon*	✓2	★★★★★	★★★☆☆
Tata Nexon	✓2	★★★★☆	★★★☆☆
Mahindra Marazzo**	✓2	★★★★☆	★★☆☆☆
Toyota Etios	✓2	★★★★☆	★★☆☆☆
Tata Zest	✓2	★★★★☆	★★☆☆☆
Suzuki Maruti Vitara Brezza	✓2	★★★★☆	★★☆☆☆
Volkswagen Polo	✓2	★★★★☆	★★★☆☆
Suzuki Marut Ertiga	✓2	★★★☆☆	★★★☆☆
Ford Aspire	✓2	★★★☆☆	★★☆☆☆
Honda Mobilio	✓2	★★★☆☆	★★☆☆☆
Renault Duster	✓1	★★★☆☆	★★☆☆☆
Suzuki Maruti Swift	✓2	★★☆☆☆	★★☆☆☆
Suzuki Maruti WagonR	✓1	★★☆☆☆	★★☆☆☆
Hyundai Santro	✓1	★★☆☆☆	★★☆☆☆
Datsun Redigo	✓1	★☆☆☆☆	★★☆☆☆
Renault Kwid (IV)	✓1	★☆☆☆☆	★☆☆☆☆
Renault Kwid (III)	✓1	☆☆☆☆☆	★★☆☆☆
Renault Kwid (III)	✕	☆☆☆☆☆	★★☆☆☆
Renault Kwid (I)	✕	☆☆☆☆☆	★★☆☆☆
Volkswagen Polo	✕	☆☆☆☆☆	★★★☆☆
Ford Figo	✕	☆☆☆☆☆	★★☆☆☆
Suzuki Maruti Eeco	✕	☆☆☆☆☆	★★☆☆☆
Hyundai Eon	✕	☆☆☆☆☆	★★☆☆☆
Suzuki Maruti Alto	✕	☆☆☆☆☆	★★☆☆☆
Renault Duster	✕	☆☆☆☆☆	★★☆☆☆
Mahindra Scorpio	✕	☆☆☆☆☆	★★☆☆☆
Renault Lodgy	✕	☆☆☆☆☆	★★☆☆☆
Datsun Go	✕	☆☆☆☆☆	★★☆☆☆
Chevrolet Enjoy	✕	☆☆☆☆☆	★★☆☆☆
Tata Zest	✕	☆☆☆☆☆	★☆☆☆☆
Suzuki Maruti Celerio	✕	☆☆☆☆☆	★☆☆☆☆
Honda Mobilio	✕	☆☆☆☆☆	★☆☆☆☆
Suzuki Maruti Swift	✕	☆☆☆☆☆	★☆☆☆☆
Hyundai i10	✕	☆☆☆☆☆	★☆☆☆☆
Tata Nano	✕	☆☆☆☆☆	☆☆☆☆☆

Fig.3 Safety rating of the car in India

Chapter 7

3D MODELLING OF THE FRONT BUMPER RECTANGULAR PIPE



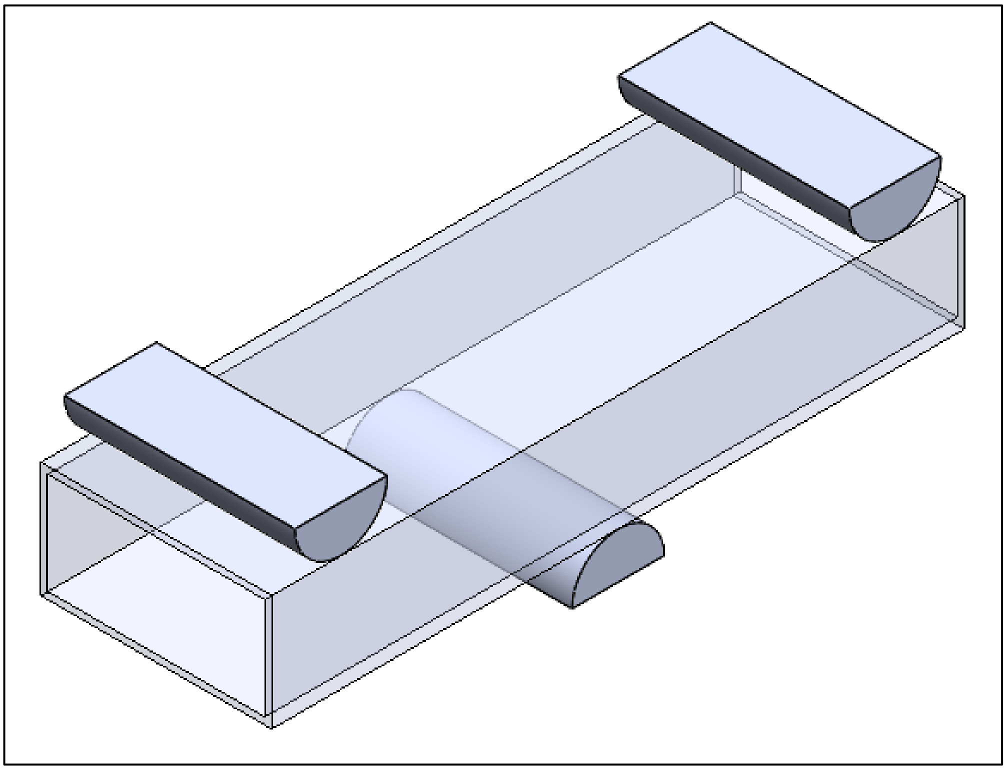


Fig.4 3D model for the FEA

Chapter 8

FINITE ELEMENT ANALYSIS

The finite element method (FEM) is a numerical problem-solving methodology commonly used across multiple engineering disciplines for numerous applications such as structural analysis, fluid flow, heat transfer, mass transport, and anything existing as a real-world force. This practice systematically yields equations and attempts to approximate the values of the unknowns. This method subdivides the overall problem into simpler sub-issues that are easier to solve. In turn, these sub-issues called finite elements require implicit vs explicit analysis.

a) CAE in the automotive industry:

CAE tools are very widely used in the automotive industry. In fact, their use has enabled the automakers to reduce product development cost and time while improving the safety, comfort, and durability of the vehicles they produce. The predictive capability of CAE tools has progressed to the point where much of the design verification is now done using computer simulations rather than physical prototype testing. CAE dependability is based upon all proper assumptions as inputs and must identify critical inputs. Even though there have been many advances in CAE, and it is widely used in the engineering field, physical testing is still used as a final confirmation for subsystems because CAE cannot predict all variables in complex assemblies (i.e. metal stretch, thinning).

b) Finite Element Method (FEM):

It is a numerical technique for finding approximate solutions to boundary value problems for partial differential equations. It is also referred to as finite element analysis (FEA). FEM subdivides a large problem into smaller, simpler, parts, called finite elements. The simple equations that model these finite elements are then assembled into a larger system of equations that models the entire problem. FEM then uses variational methods from the calculus of variations to approximate a solution by minimizing an associated error function.

c) Ansys Meshing:

Over the last 20 years, Meshing has evolved into the leading premier pre-processor for concept and high-fidelity modeling. The advanced geometry and meshing capabilities provide an environment for rapid model generation.

The ability to generate high quality mesh quickly is one of Ansys Workbench core competencies. Industry trends show a migration to modular sub-system design and continued exploration of new materials; Ansys Workbench has advanced model assembly tools capable of supporting complex sub-system generation and assembly, in addition, modeling of laminate composites is supported by advanced creation, editing and visualization tools. Design change is made possible via mesh morphing and geometry dimensioning. Ansys Workbench is a solver neutral environment that also has an extensive API which allows for advanced levels customization.

Analysis of Original Brake pedal

The Finite Element Method is a numerical approximation method, in which the complex structure is divided into number of small parts that is pieces and these small parts are called as finite elements. These small elements are connected to each other by means of small points called as nodes. As the finite element method uses matrix algebra to solve the simultaneous equations, so it is also known as structural analysis and it's becoming primary analysis tool for designers and analysts.

The three basic FEA process are

a) Preprocessing phase

b) Processing or solution phase

c) Post processing phase

A static structural analysis is the analysis displacements, stresses, strains and forces on structure or a component due to load application. The structures response and loads are assumed to vary slowly with respect to time. There are various types of loading that can be applied in this analysis which are externally applied forces and pressures, and temperatures.

FEA Pre Processing

The pre-processing of the connecting rod is down for the purpose of the dividing the problem into nodes and elements, developing equation for an element, applying boundary conditions, initial conditions and for applying loads. The information required for the pre-processing stage of the connecting rod is as follows

• **Constraints**: The nodes around the connecting rod holes have a rigid element connecting them to the center of the hole which has of its degree of freedom fixed. The element which is used to fix connecting rod and crank shaft is fixed and used as a rigid element.

The minimum and maximum are set, together with other mesh parameters such as element type and material. The selected object is ready for further analysis.

Post Processing

The acceptability of the design of the connecting rod needs to be considered from the results of the analysis. The guidance for the modification of the rod need to be available if the design is not considered to be acceptable for the connecting rod are as follows.

Model acceptance criteria: the maximum von-Mises stress must be less than the material yield strength for the duration of the component. The deflection is considered and the maximum Von-Mises stress must be less than the yield strength for abuse load case.

The finite element method (FEM), is a numerical method for solving problems of engineering and mathematical physics. Typical problem areas of interest include structural analysis, heat transfer, fluid flow, mass transport, and electromagnetic potential. The analytical solution of these problems generally require the solution to boundary value problems for partial differential equations. The finite element method formulation of the problem results in a system of algebraic equations. The method yields approximate values of the unknowns at discrete number of points over the domain. To solve the problem, it subdivides a large problem into smaller, simpler parts that are called finite elements.

In the first step, the element equations are simple equations that locally approximate the original complex equations to be studied, where the original equations are often partial differential equations (PDE). The process, in mathematical language, is to construct an integral of the inner product of the residual and the weight functions and set the integral to zero. In simple terms, it is a procedure that minimizes the error of approximation by fitting trial functions into the PDE. The residual is the error caused by the trial functions, and the weight functions are polynomial approximation functions that project the residual. The process eliminates all the spatial derivatives from the PDE, thus approximating the PDE locally with

- A set of algebraic equations for steady state problems,
- A set of ordinary differential equations for transient problems.

These equation sets are the element equations. They are linear if the underlying PDE is linear, and vice versa. Algebraic equation sets that arise in the steady state problems are solved using numerical linear algebra methods, while ordinary differential equation sets that arise in the transient problems are solved by numerical integration using standard techniques such as Euler's method or the Runge-Kutta method.

FEM is best understood from its practical application, known as finite element analysis (FEA). FEA as applied in engineering is a computational tool for performing engineering analysis. It includes the use of mesh generation techniques for dividing a complex problem into small elements, as well as the use of software program coded with FEM algorithm. In applying FEA, the complex problem is usually a physical system with the underlying physics such as the Euler-Bernoulli beam equation, the heat equation, or the Navier-Stokes equations expressed in either PDE or integral equations, while the divided small elements of the complex problem represent different areas in the physical system.

In present research for analysis ANSYS (**An**alysis **Sys**tem) software is used. Basically, its present FEM method to solve any problem. Following are steps in detail

1. Geometry
2. Discretization (Meshing)
3. Boundary condition
4. Solve (Solution)
5. Interpretation of results

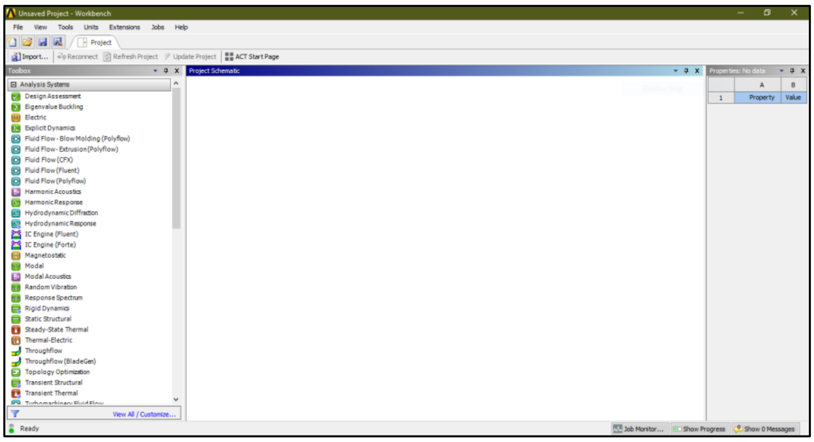


Fig. Workbench Menu

Workbench contain analysis of different types namely static, modal, harmonic, explicit dynamics, CFD, ACP tool post, CFX, topology optimization etc. as per problem defined.

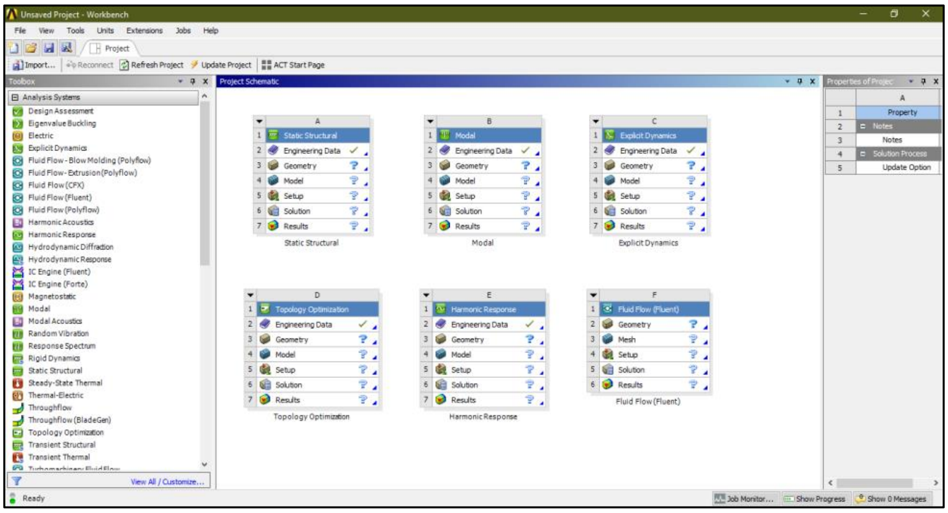


Fig Workbench Various Options

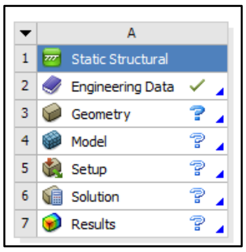


Fig Workbench Option A

Step 1: Details of material namely copper, steel, grey cast iron, composite material, fluid domain material is defined in engineering data. i.e. ANSYS default material is structural steel.

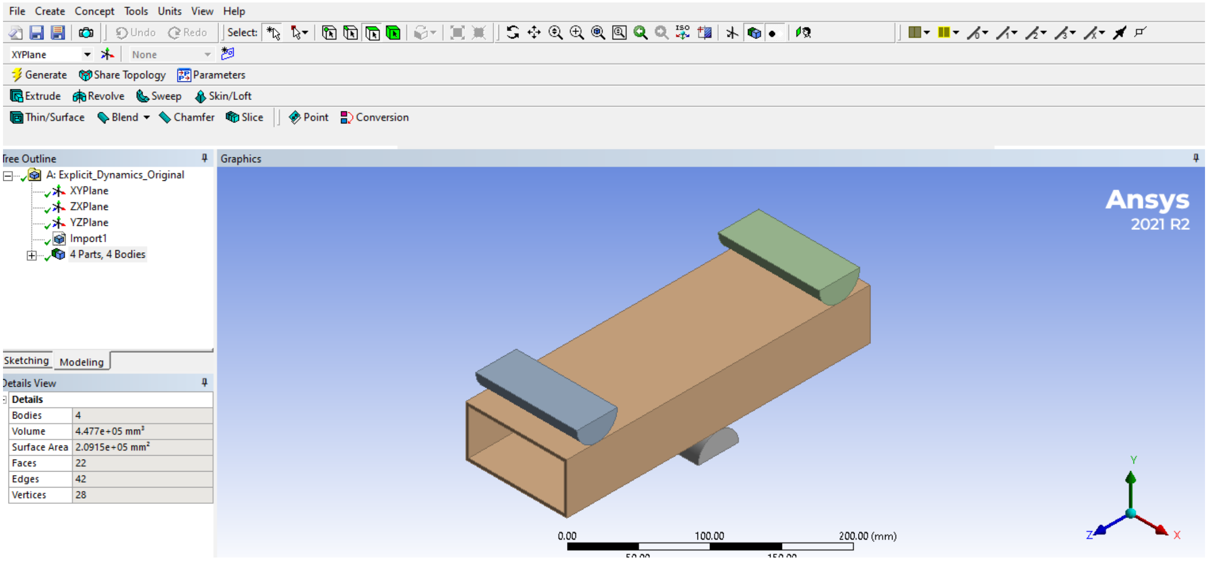
Step 2: Import of geometry created in any CAD software namely CATIA, PRO E, SOLIDWORK, INVENTOR etc. in geometry section. If any correction is to be made it can be created in geometry section in Design modeller or space claim.

Step 3: In model section after import of component

- Material is assigned to component as per existing material
- Connection is checked in contact region i.e. bonded, frictionless, frictional, no separation etc. for multi body components.
- Meshing or discretization is performed i.e. to break components in small pieces (elements) as per size i.e. preferably tetra mesh and hexahedral mesh for 3D geometry and for 2 D quad or tria are generally preferred.

Step 4: Boundary condition are applied as per analysis namely in fixed support, pressure, force, displacement, velocity as per condition.

Step 5: Now problem is well defined and solve option is selected to obtain the solution in the form of equivalent stress, strain, energy, reaction force etc.



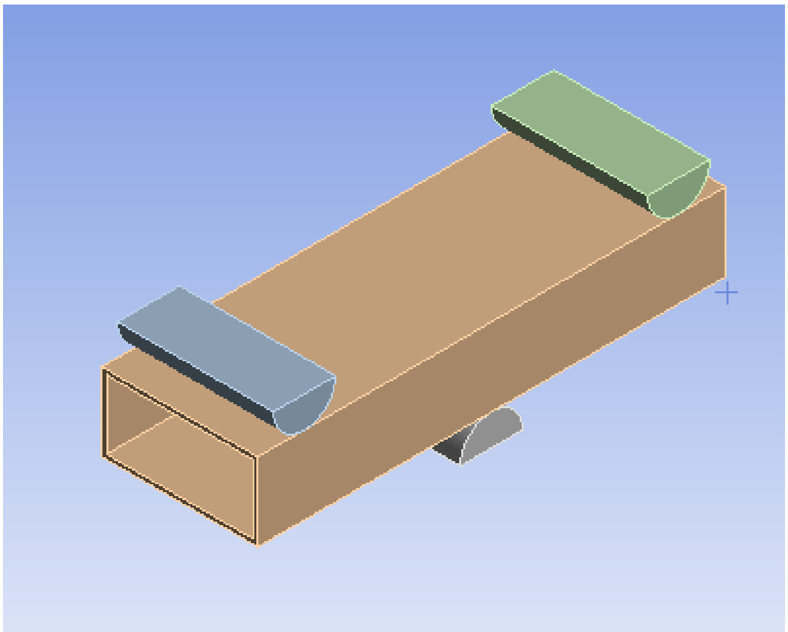


Fig. 5 Geometry of Bumper

Properties of Outline Row 4: Structural Steel NL

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	7850	kg m^-3		
4	Isotropic Elasticity				
10	Bilinear Isotropic Hardening				
11	Yield Strength	2.5E+08	Pa		
12	Tangent Modulus	1.45E+09	Pa		
13	Specific Heat Constant Pressure, C_p	434	J kg^-...		

Fig. 6 Material properties of Bumper

Chapter 9

MESH

ANSYS Meshing is a general-purpose, intelligent, automated high-performance product. It produces the most appropriate mesh for accurate, efficient Multiphysics solutions. A mesh well suited for a specific analysis can be generated with a single mouse click for all parts in a model. Full controls over the options used to generate the mesh are available for the expert user who wants to fine-tune it. The power of parallel processing is automatically used to reduce the time you have to wait for mesh generation.

Creating the most appropriate mesh is the foundation of engineering simulations. ANSYS Meshing is aware of the type of solutions that will be used in the project and has the appropriate criteria to create the best suited mesh. ANSYS Meshing is automatically integrated with each solver within the ANSYS Workbench environment. For a quick analysis or for the new and infrequent user, a usable mesh can be created with one click of the mouse. ANSYS Meshing chooses the most appropriate options based on the analysis type and the geometry of the model. Especially convenient is the ability of ANSYS Meshing to automatically take advantage of the available cores in the computer to use parallel processing and thus significantly reduce the time to create a mesh. Parallel meshing is available without any additional cost or license requirements.

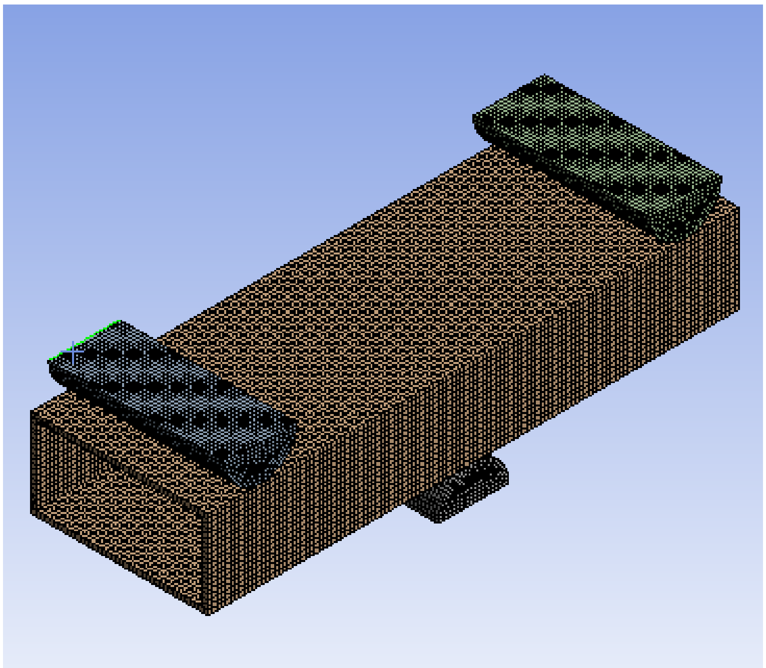
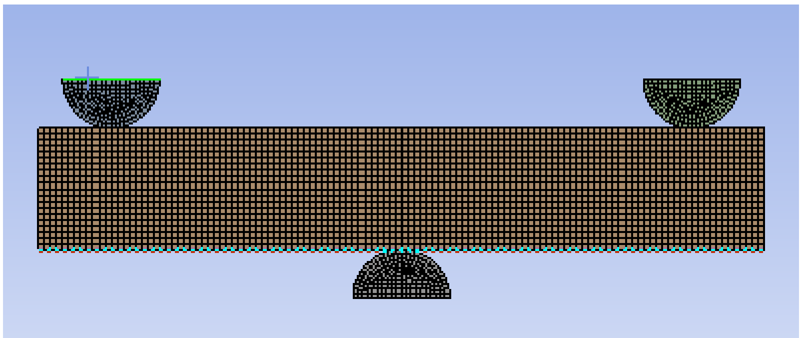
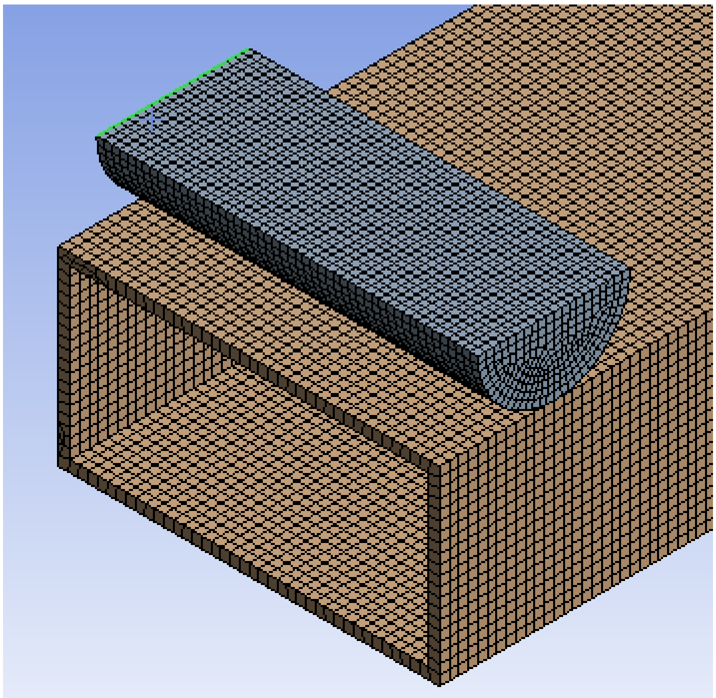


Fig. 7 Meshing details





Statistics	
Nodes	64498
Elements	46030

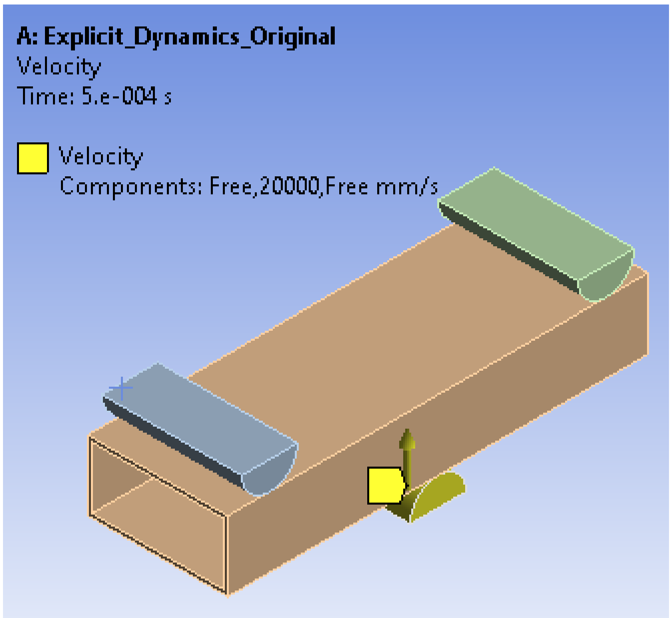
Fig. 7 Meshing details

Chapter 10

Boundary Condition

A boundary condition for the model is that the setting of a well-known value for a displacement or an associated load. For a specific node you'll be able to set either the load or the displacement but not each.

The main kinds of loading obtainable in FEA include force, pressure and temperature. These may be applied to points, surfaces, edges, nodes and components or remotely offset from a feature.



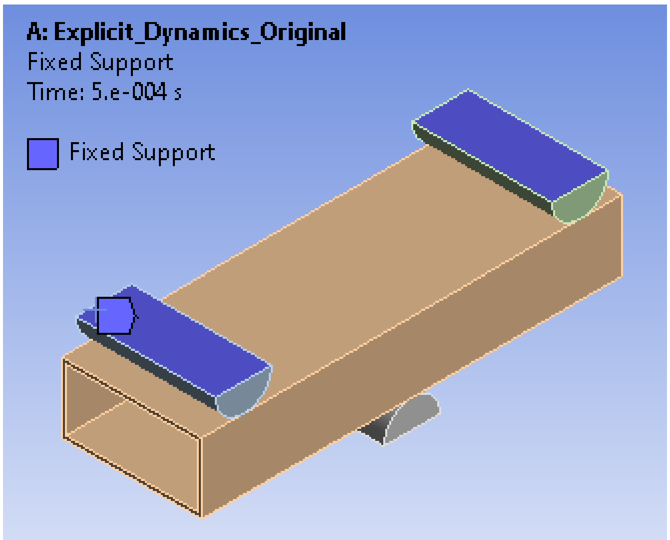
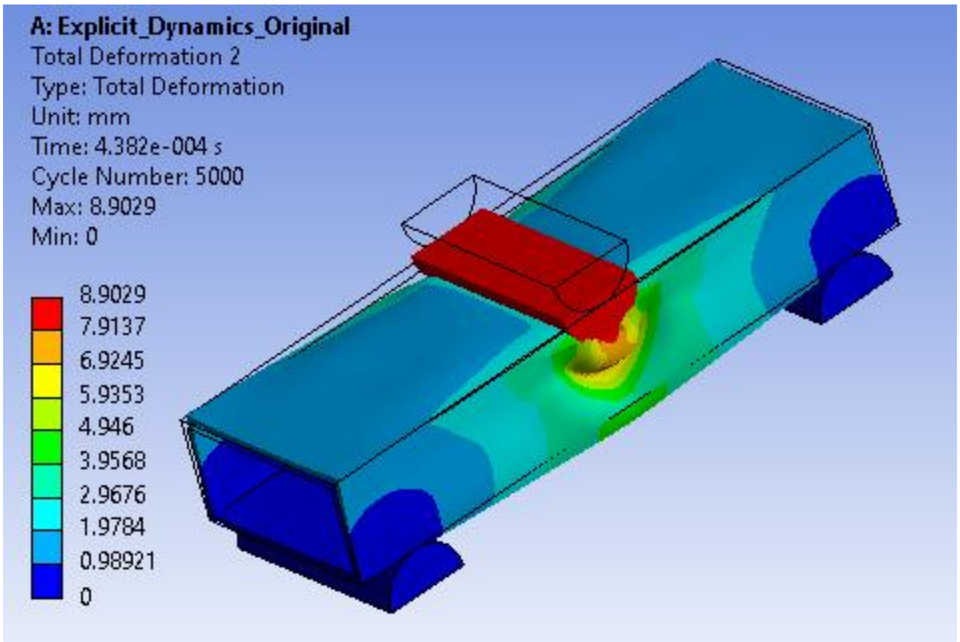


Fig. 8 Velocity and Fixed support

Chapter 11

RESULTS



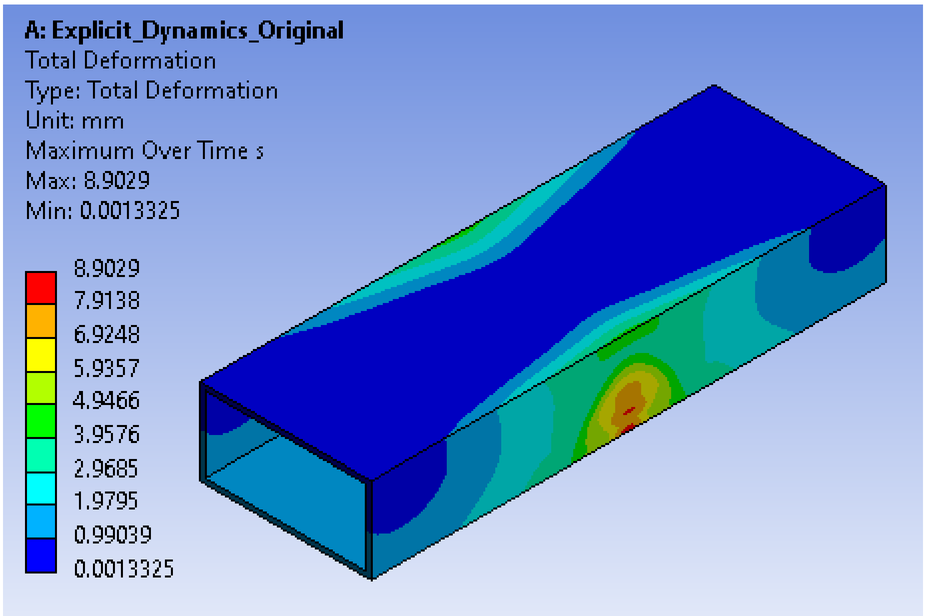


Fig. 9 Total deformation on front bumper pipe

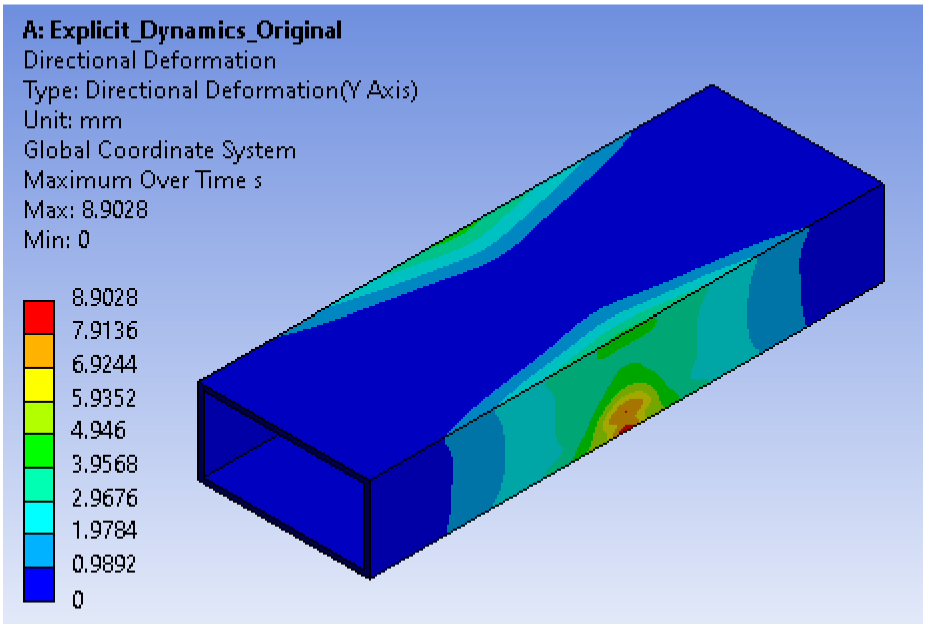


Fig. Direction deformation at y direction

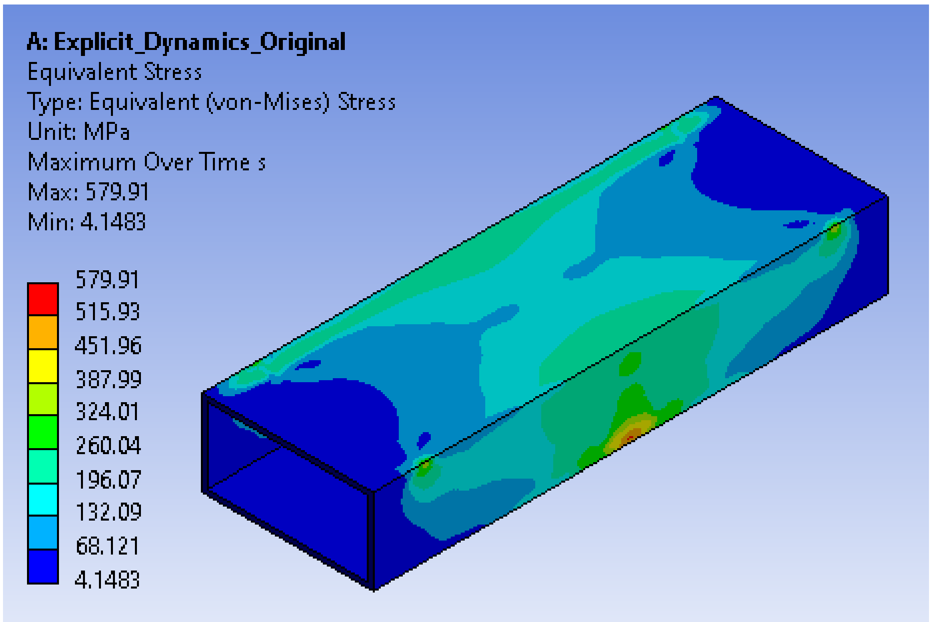
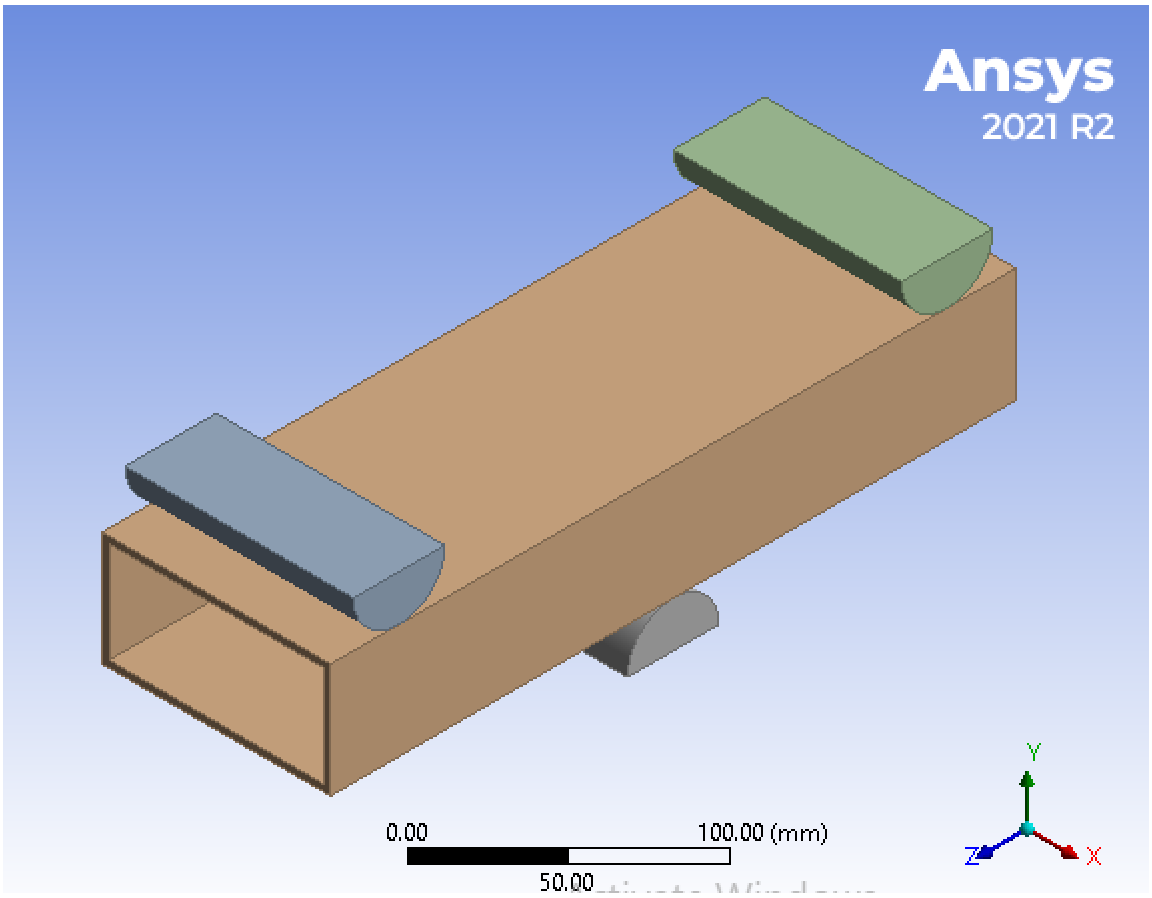


Fig. Equivalent stress generated on the front bumper pipe



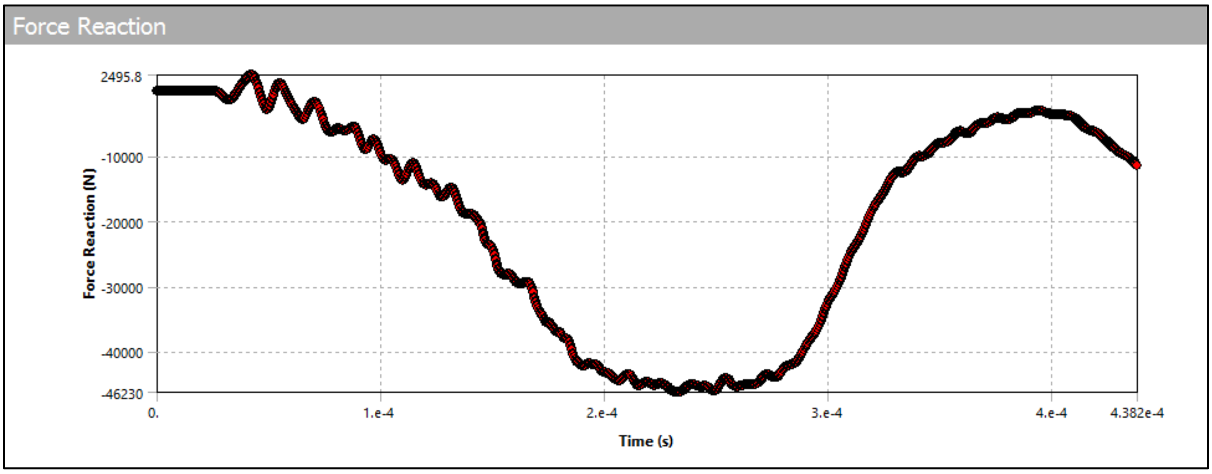
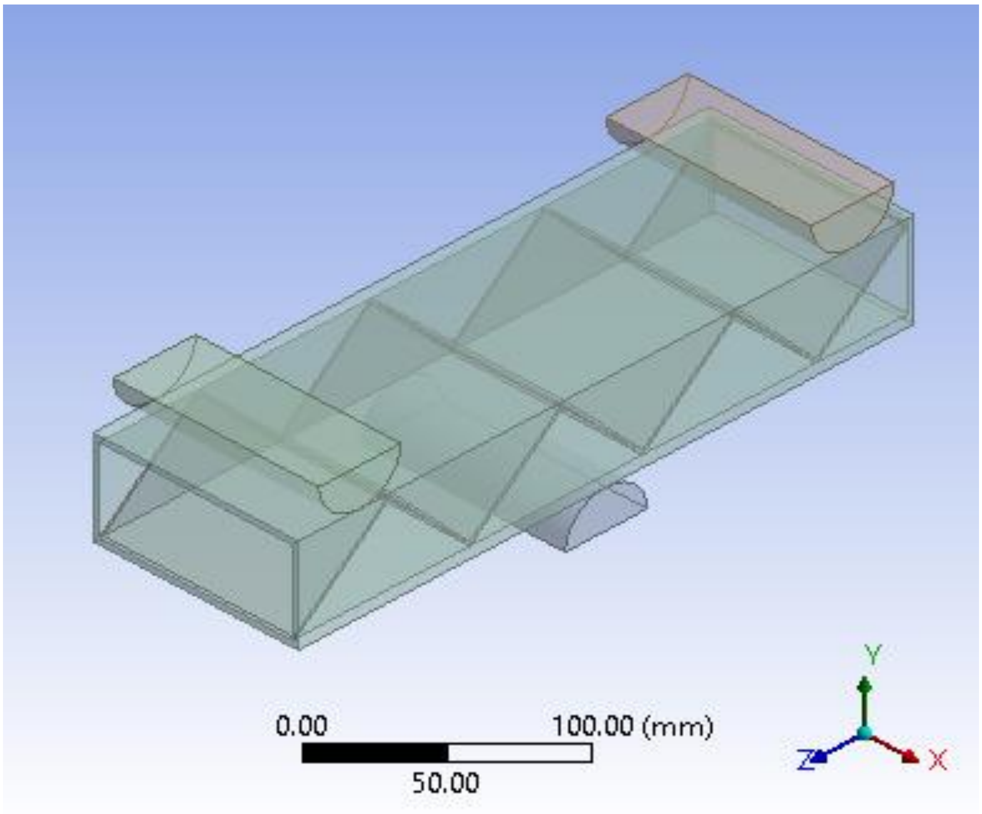


Fig. 10 Reaction force

Results	
Minimum	-46230 N
Maximum	2495.8 N
Filter	

Fig. Values of reaction force

Zig zag Shape PLA



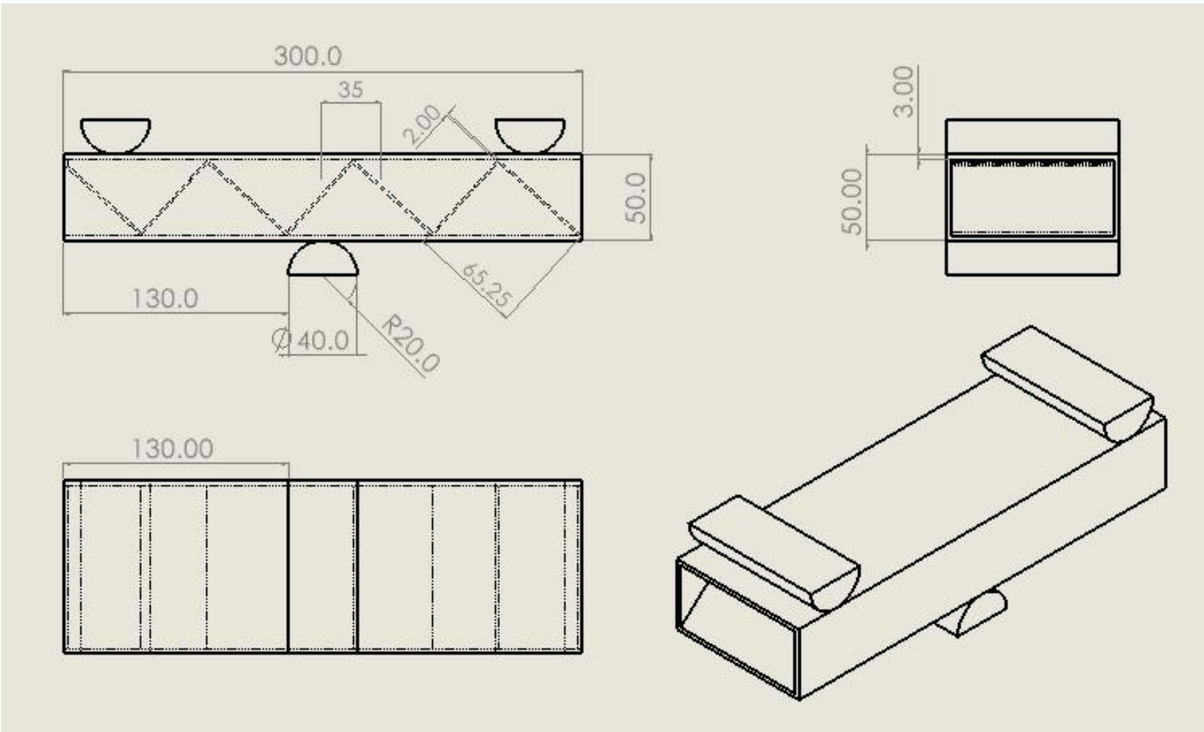


Fig. 11 Geometry of Zig zag Shape PLA

Properties of Outline Row 4: PLA NL

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	1240	kg m^-3		
4	Isotropic Elasticity				
5	Derive from	Young's Modulu...			
6	Young's Modulus	1280	MPa		
7	Poisson's Ratio	0.3			
8	Bulk Modulus	1.0667E+09	Pa		
9	Shear Modulus	4.9231E+08	Pa		
10	Bilinear Isotropic Hardening				
11	Yield Strength	26.082	MPa		
12	Tangent Modulus	400	MPa		

Fig. 12 Material Properties for PLA

Properties	
Volume	5.0352e+005 mm^3
Mass	3.5836 kg

Fig. Weight of PLA Material

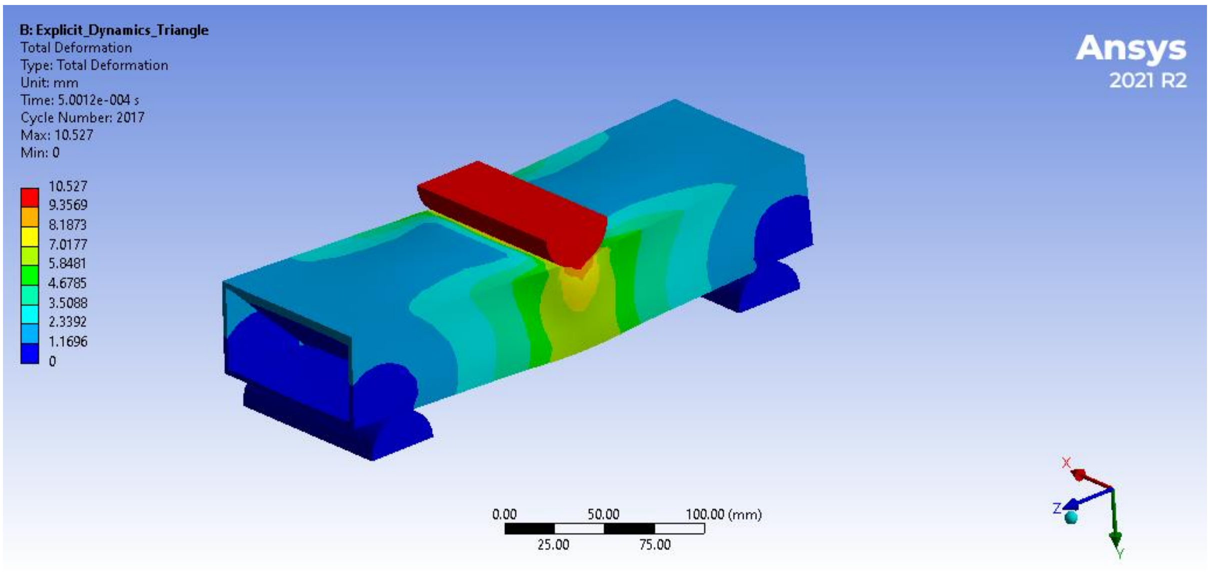


Fig. Total deformation

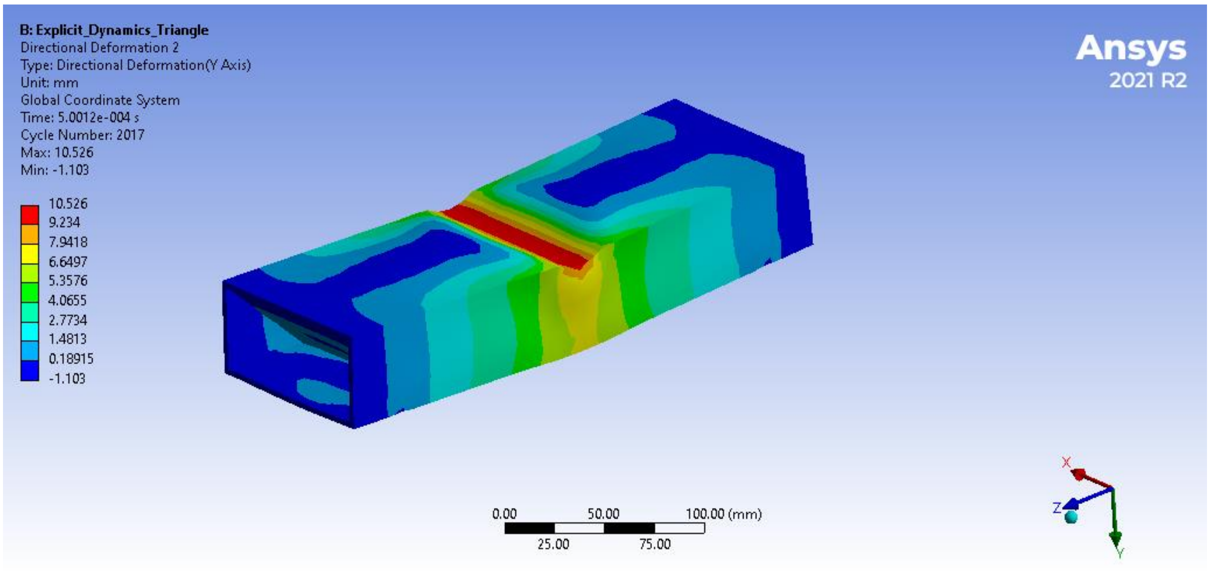


Fig . 13 Total Deformation in Bumper

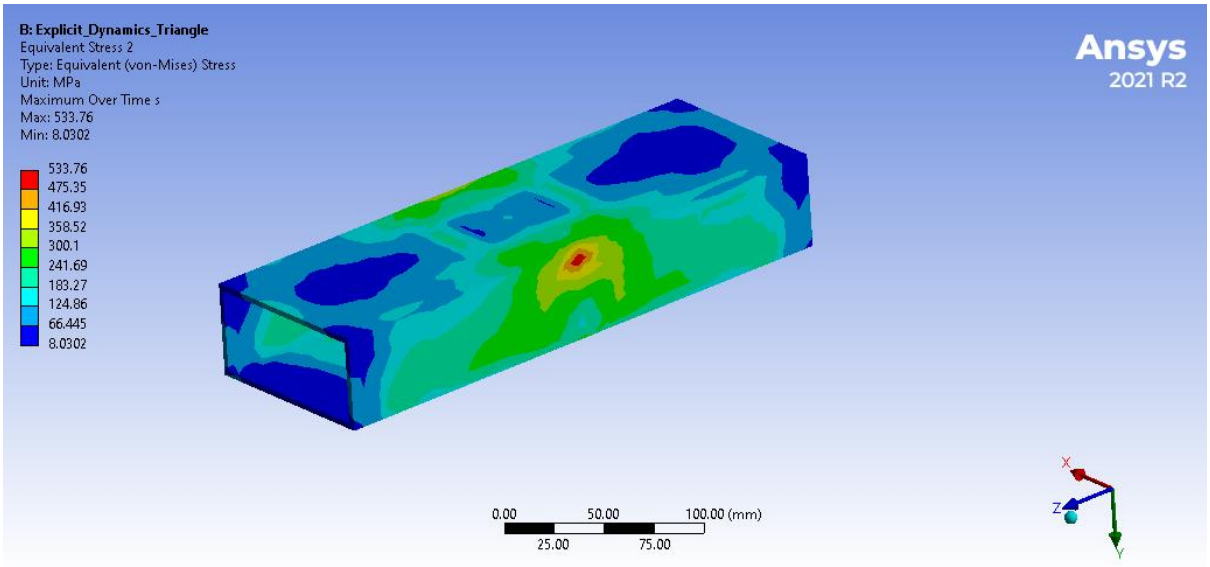


Fig. Equivalent stress generated on the front bumper pipe

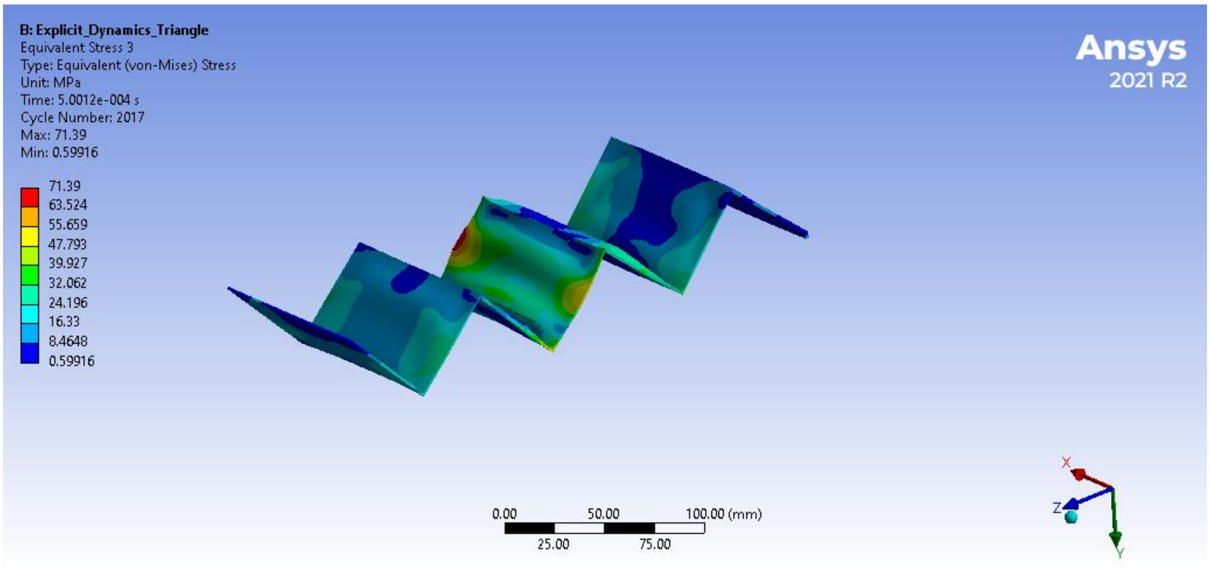
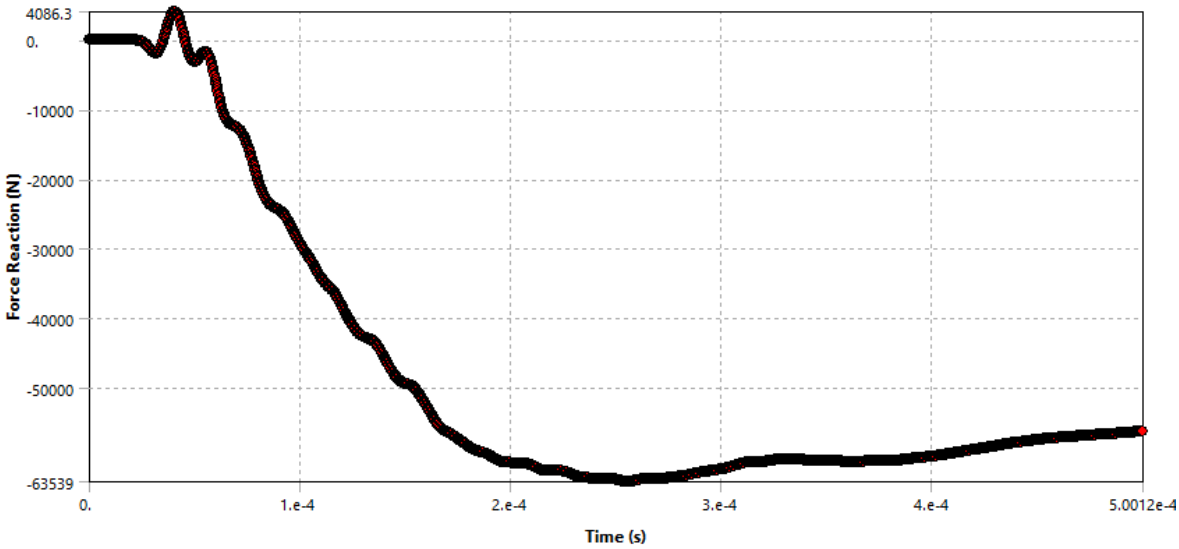


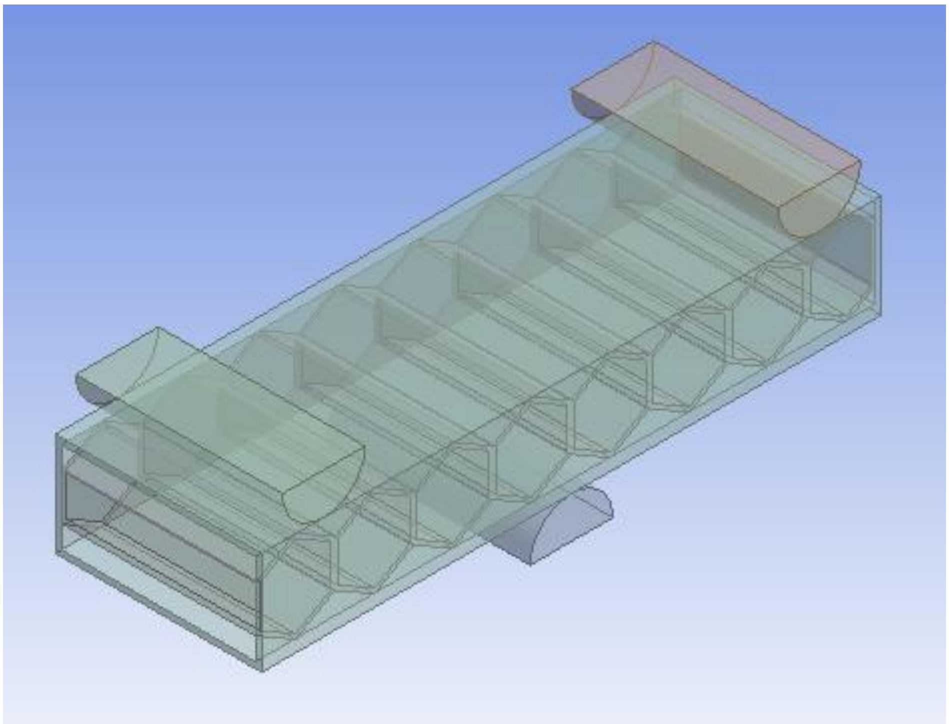
Fig. Equivalent stress generated on the front bumper Zig Zag Shape



Results	
Minimum	-63539 N
Maximum	4086.3 N

Fig. 14 Reaction force

Honeycomb Shape PLA



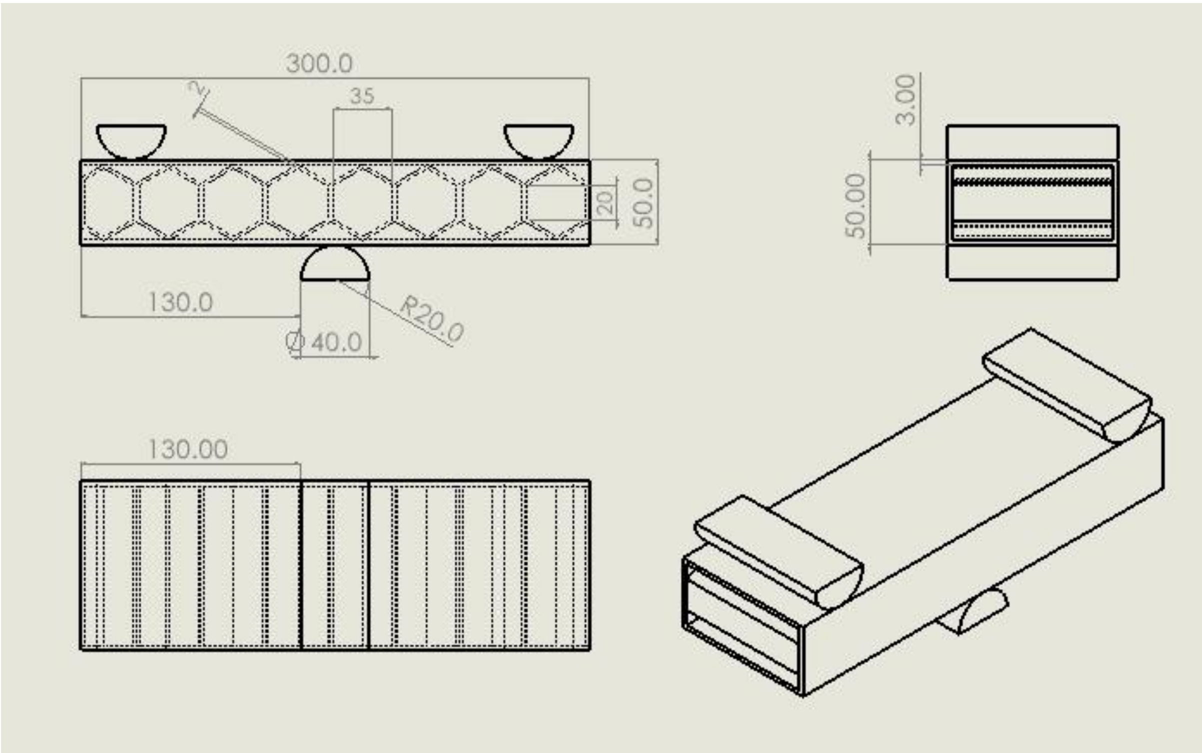


Fig. 15 Geometry of Honeycomb Shape PLA

Properties of Outline Row 4: PLA NL

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	1240	kg m^-3		
4	Isotropic Elasticity				
5	Derive from	Young's Modulu...			
6	Young's Modulus	1280	MPa		
7	Poisson's Ratio	0.3			
8	Bulk Modulus	1.0667E+09	Pa		
9	Shear Modulus	4.9231E+08	Pa		
10	Bilinear Isotropic Hardening				
11	Yield Strength	26.082	MPa		
12	Tangent Modulus	400	MPa		

Fig. Material Properties for PLA

Properties

Volume	6.2997e+005 mm^3
Mass	3.7404 kg

Fig. Weight of PLA Material

Fig. 16 Material Properties for PLA used for Honeycomb shape

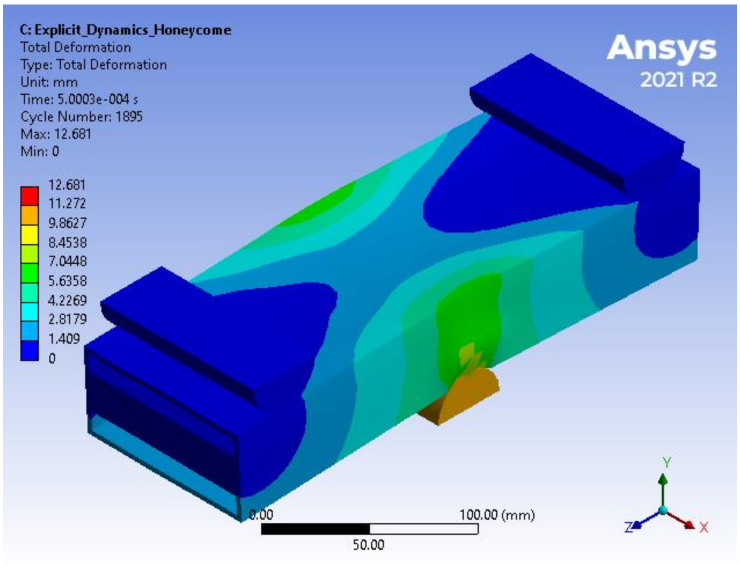


Fig. 17 Total deformation

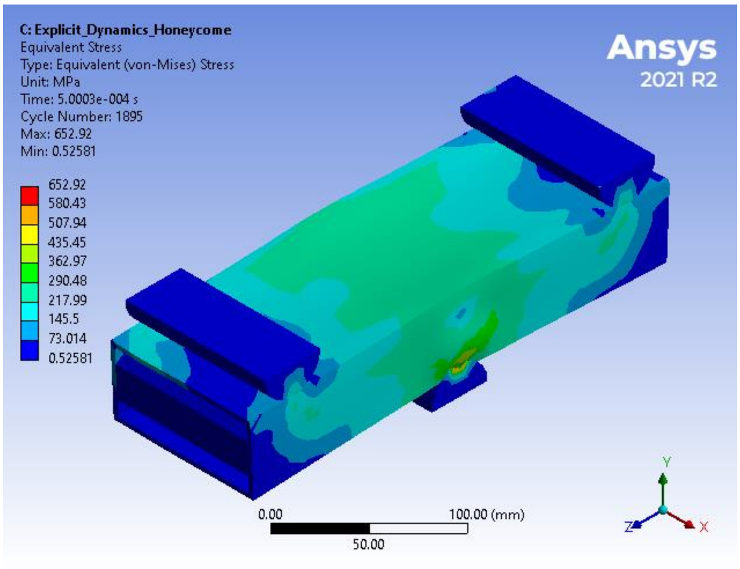


Fig. Equivalent stress generated on the front bumper pipe

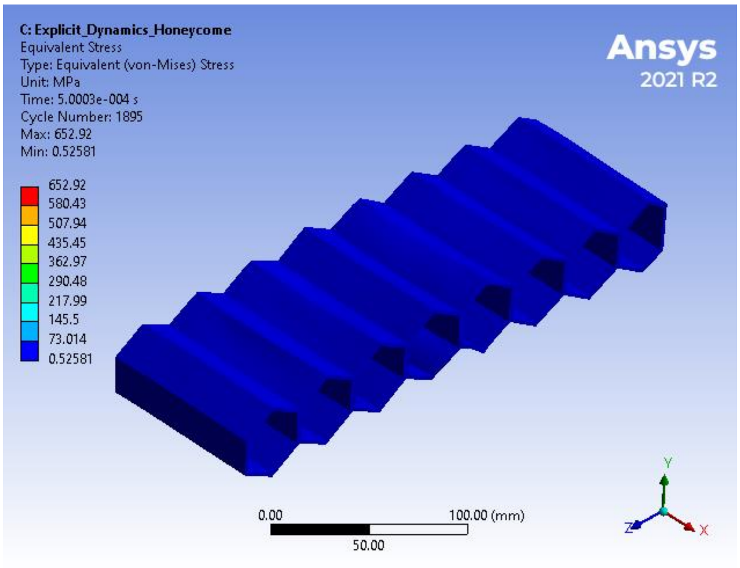
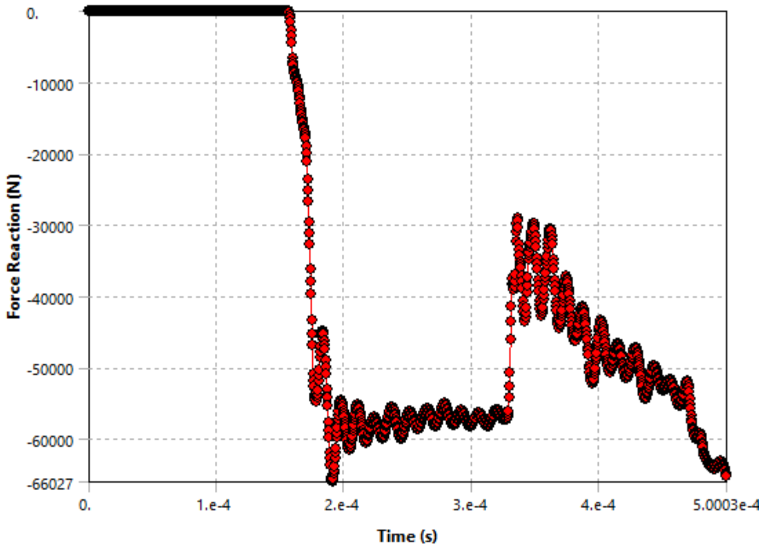


Fig. Equivalent stress generated on the front bumper Honeycomb Shape



Results	
Minimum	-66027 N
Maximum	0. N

Fig. 18 Reaction force

Chapter 12

EXPERIMENTAL TEST

A Universal Testing Machine (UTM) is used to test both the tensile and compressive strength of materials. Universal Testing Machines are named as such because they can perform many different varieties of tests on an equally diverse range of materials, components, and structures.

Universal Testing Machines can accommodate many kinds of materials, ranging from hard samples, such as metals and concrete, to flexible samples, such as rubber and textiles. This diversity makes the Universal Testing Machine equally applicable to virtually any manufacturing industry.

The UTM is a versatile and valuable piece of testing equipment that can evaluate materials properties such as tensile strength, elasticity, compression, yield strength, elastic and plastic deformation, bend compression, and strain hardening. Different models of Universal Testing Machines have different load capacities, some as low as 5kN and others as high as 2,000kN.

Sr. No.	**Description**	**Information with Unit**
1	Max Capacity	400KN
2	Measuring range	0-400KN
3	Least Count	0.04KN
4	Clearance for Tensile Test	50-700 mm
5	Clearance for Compression Test	0- 700 mm
6	Clearance Between column	500 mm
7	Ram stroke	200 mm
8	Power supply	3 Phase , 440Volts , 50 cycle. A.C
9	Overall dimension of machine (L*W*H)	2100*800*2060
10	Weight	2300Kg

Table for Specification of UTM

- Fixture is manufactured according to component designed.
- Single force is applied as per FEA analysis and reanalysis is performed to determine strain by numerical and experimental testing.
- Strain gauge is applied as per FEA results to maximum strained region and during experimental testing force is applied as per numerical analysis to check the strain obtained by numerical and experimental results.
- During strain gage experiment two wires connected to strain gage is connected to micro controller through the data acquisition system and DAQ is connected to laptop. Strain gage values are displayed on laptop using DEWESOFT software.

Metal Hybrid 100 x 50

Load(kN)	Deflection (mm)	
10	0.1	
14	0.3	
19	0.6	
26	1.1	
33	2.1	
40	2.6	
43.5	3.2	
44.2	3.6	
44.9	4.1	MAX LOAD

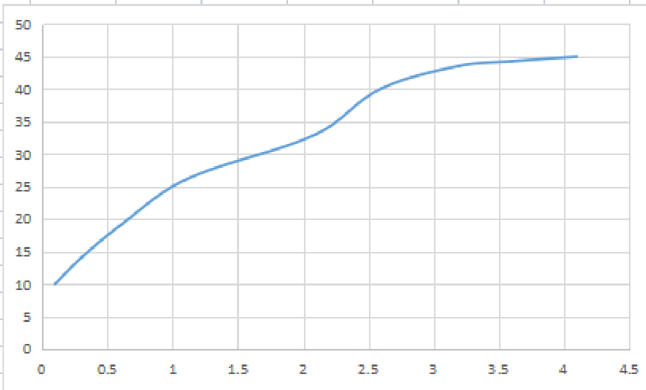
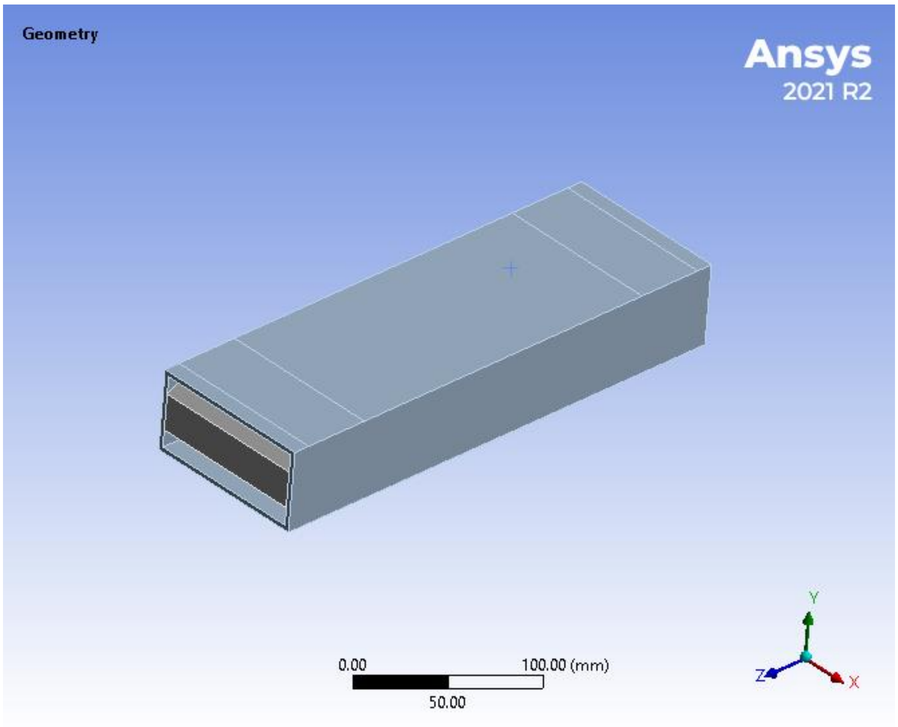
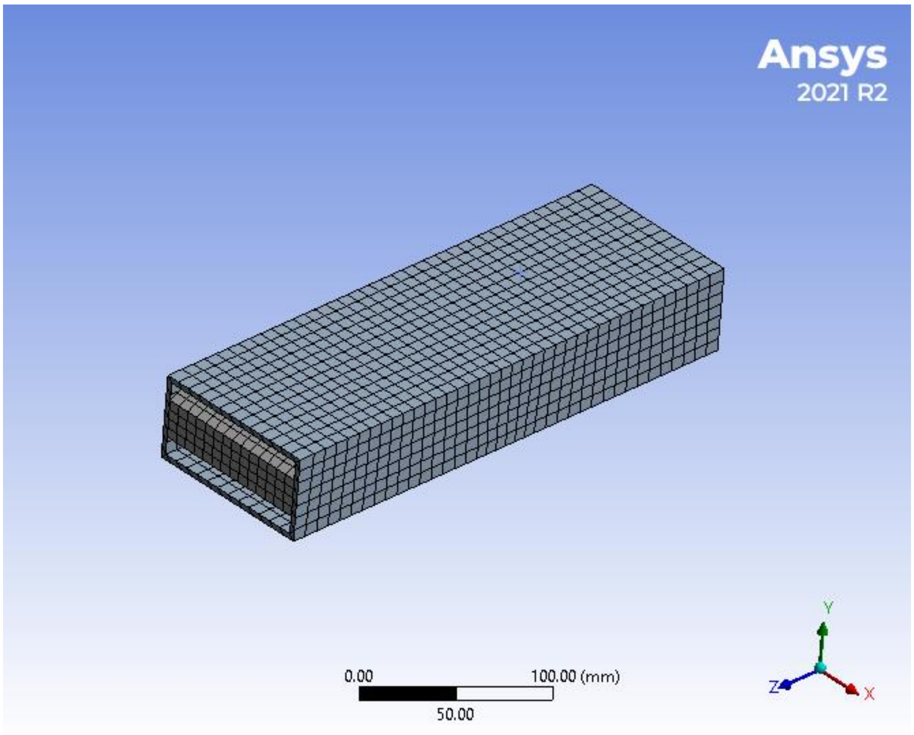


Fig. DEWESOFT Software Graphical Representation

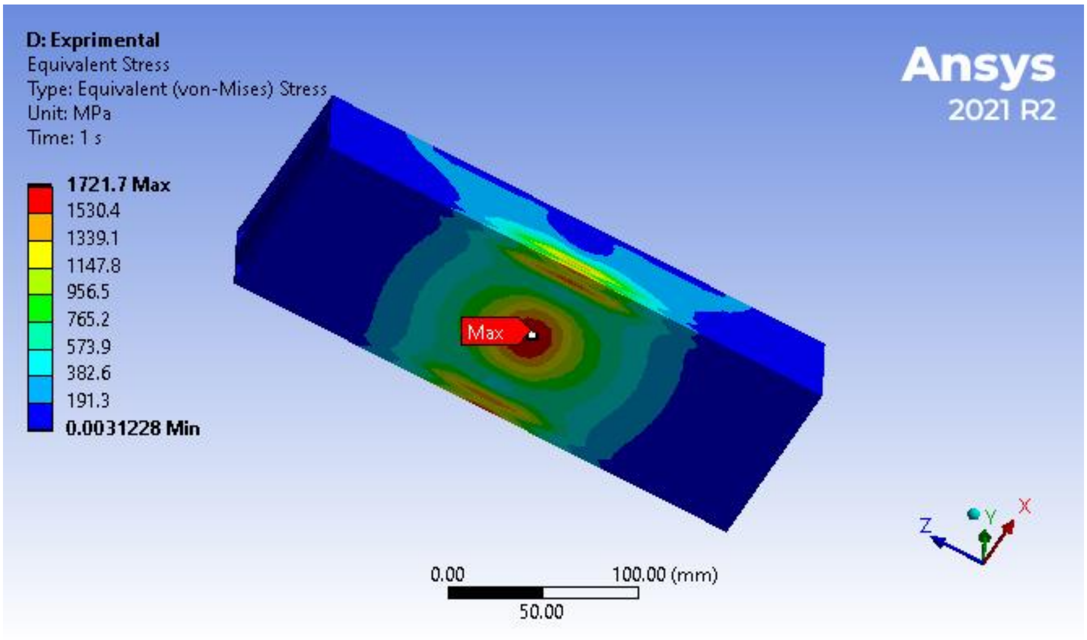
Chapter 13

EXPERIMENTAL FEA





Statistics	
Nodes	25302
Elements	3938



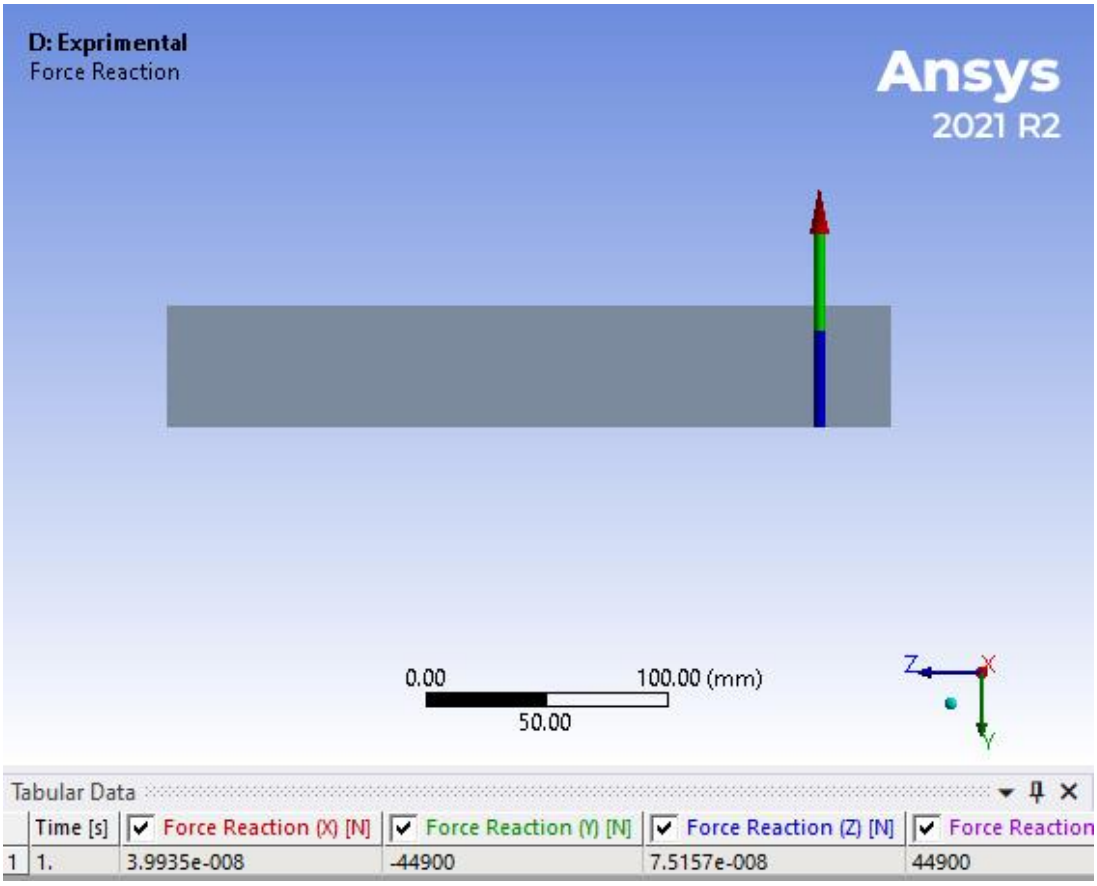


Fig. Load applied

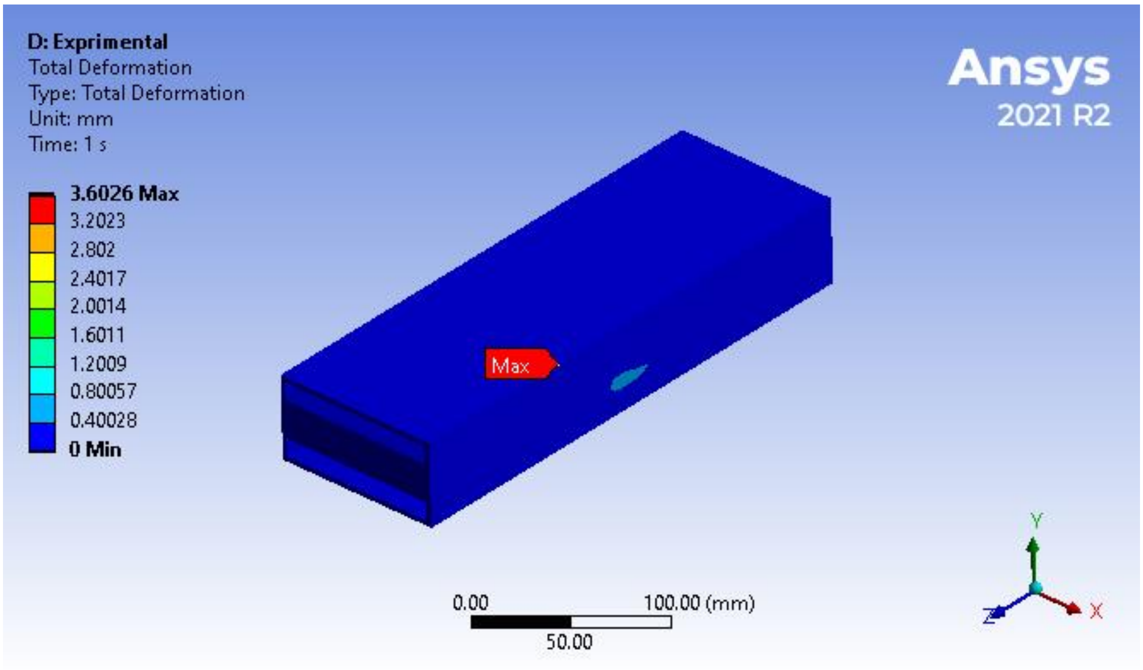


Fig. Deformation generated

Fig. 19 Experimental FEA

Chapter 14

CONCLUSION

- In this project, study on the 3D printing technology and pipes use for the front bumper for impact improvisation.
- Developed the literature survey on the bumper and metal plastic hybrid.
- Developed the 3D cad model of front bumper rectangular pipe using CATIA software.
- Performed the explicit dynamic analysis on the front bumper pipe using ANSYS software and find out the reaction force of the bumper without plastic stiffeners.
- The reaction force of the existing bumper is 46230 N and the negative direction shows the direction.
- The equivalent stress and total deformation observed on the model is 580 MPa and 8.9 mm
- As per dynamic analysis the force reaction observed on the existing front bumper pipe is 46230N. To increase the shock absorbing capacity of the bumper add zig zag structure into the pipe.
- As per Experimental testing after applying the load of the 44.9K the deformation observed is 4.1 mm in the pipe.
- As per static analysis we applied the load on the same load as experimental testing and check out the deformation to validate the result.
- The deformation observed in the FEA is 3.6 mm.

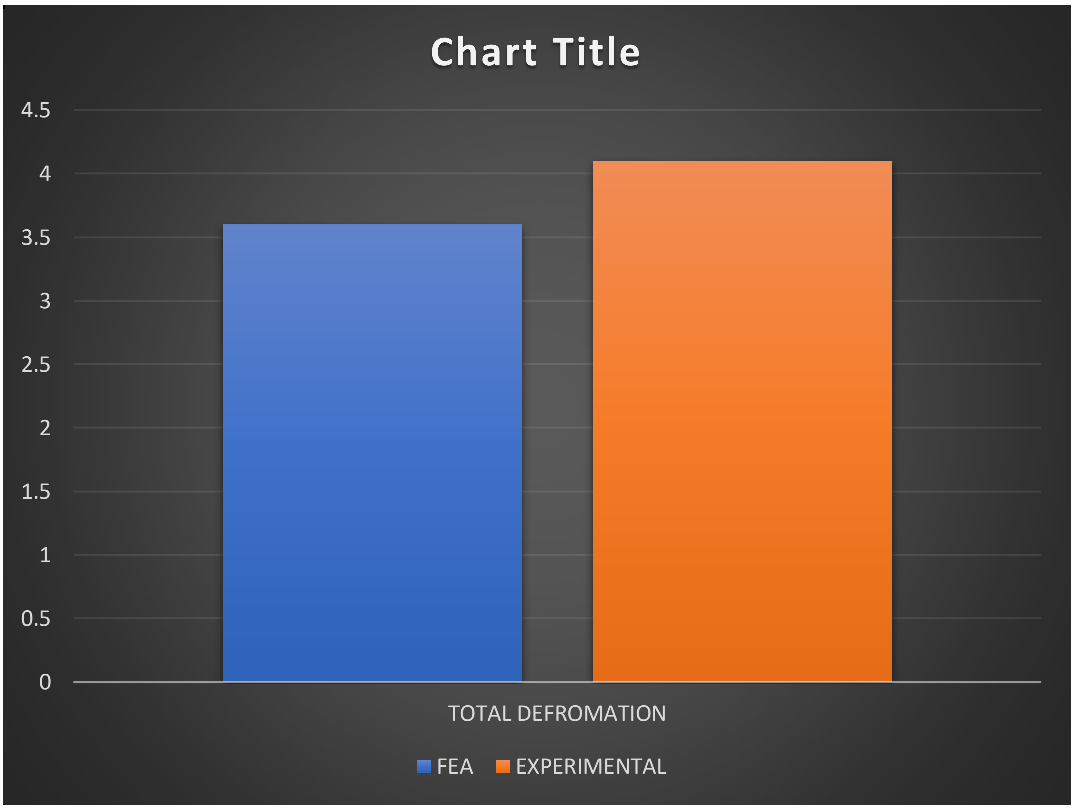


Fig. 20 Total deformation comparison

Chapter 15

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